Detecting Text-Reading Hallucinations in Vision-Language Descriptions of Cyrillic Signage, Packaging and Documents for Blind Users

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Abstract

Purpose: Blind people increasingly rely on vision-language models to read signs, product labels and documents aloud, but a model that cannot read a word tends to produce a plausible one instead. We measure how often this happens for Ukrainian (Cyrillic) text and whether it can be detected at run time.

Methods: Four Claude models of one vendor were evaluated. Study 1 uses 567 synthetic images in a factorial design that crosses image quality, framing (crop, occlusion) and the type of the distinctive name (real word, invented word, near-word). Study 2 uses 176 real images, Open Food Facts packaging photographs and official Ukrainian documents, with model-independent ground truth. A paired near-word probe, repeated runs, paired tests and a two-rater automated audit of the labeller complete the design.

Results: Only 2.8–4.1% (Study 1) and 3.9–5.6% (Study 2) of quoted words were hallucinated; after the audit, 2.5–3.6% and 2.2–3.1%. Yet 35–46% and 58–73% of

descriptions contained at least one word labelled as hallucinated, mostly without hedging. On real documents, a printed near-word was reported as the real word it resembles in 44–56% of pages. Combining the model's own re-reading, an independent OCR pass and agreement between models detected hallucinated words with an AUROC of 0.95–0.97 (Study 1) and 0.86–0.90 (Study 2).
Conclusion: Unhedged text-reading hallucinations are rare per word but common per description, and they can be flagged at run time, so assistive reading applications can tell blind users which words to doubt. The findings are scoped to one model family.

Keywords: blind and visually impaired users, vision-language models, hallucination detection, scene text reading, Cyrillic script, assistive technology

1 Introduction

Reading text is one of the most frequent reasons blind people photograph their surroundings. The first crowd-powered visual question answering service, VizWiz, already found that a large share of questions asked what a label, sign or screen said [1], and later analyses of tens of thousands of such questions confirmed that reading and product identification dominate everyday visual needs [2, 3]. Dedicated text readers such as Seeing AI and OrCam serve this need [4, 5]. Since 2023, general-purpose vision-language models (VLMs) have increasingly taken over the task. Scene-description apps built on these models let a blind user point a phone at a medicine box or a utility bill and receive a fluent spoken description [6–8].

Fluency is the problem. A conventional OCR engine that cannot read a word usually produces visible garbage, and a sighted user would notice it. A VLM that cannot read a word tends to produce a *plausible* word instead: a common brand name in place of an unusual one, a complete street name where the sign is cut off by the edge of the frame, or a date that is one digit off. For a blind user the description is often the only access to the text. Blind users are known to place considerable trust in machine-generated descriptions even when they are wrong [9], and they have few independent ways to check them [10, 11]. A hallucinated expiry date, dosage or amount due is therefore not a cosmetic error but an accessibility failure with practical consequences.

This risk is plausibly greater for Cyrillic text, and for Ukrainian in particular. Text-centric benchmarks report clearly lower VLM accuracy for non-Latin scripts [12, 13]. Ukrainian adds letters that do not exist in Russian (і, ї, є, ґ) and an apostrophe inside words, so a model with a strong Russian or generic Cyrillic prior can silently "correct" them. Ukrainian everyday text is also dense with proper names, such as brands, surnames, street names and invented shop names, which a language prior cannot predict.

This paper asks two questions:

RQ1 When a VLM describes a photograph of Cyrillic signage, packaging or a document for a blind user, how often does it assert text that is not printed there, and what drives these text-reading hallucinations?

RQ2 Can such hallucinations be detected automatically, without ground truth, using signals that an assistive application could compute at run time?

Answering RQ1 requires knowing exactly what is printed and what is actually visible in the photograph. We address it in two studies. Study 1 is a controlled factorial experiment on 567 synthetic, photograph-like images of Ukrainian signs, labels and documents. Every character and its visibility are known, and the design crosses image quality, framing (cropping by the frame edge, finger or glare occlusion) and the type of the distinctive name: a real word, an invented word, or a *near-word* one letter away from a real word. Study 2 repeats the measurements on 176 images of real Ukrainian text. These are 108 packaging photographs from Open Food Facts and 68 official and public documents, such as Pension Fund and city-council forms, Ministry of Health medical forms, utility bills, bank forms, medicine leaflets and notices. Each has a ground truth that is independent of the evaluated models, and each is degraded and framed in the same way as in Study 1. A paired near-word probe edits one word on real document pages.

We evaluated four Claude models of one vendor (Haiku 5.5, Sonnet 5.5, Opus 5.5 and Fable 5.1) with a blind-user description prompt and a constrained transcription prompt. Every word the models quote as printed text was labelled automatically against the ground truth. The labeller was checked by two independent automated raters. The study is a comparison within one model family; it makes no claim about vision-language models in general.

The contributions are:

1. a synthetic factorial benchmark and a real-image benchmark of Ukrainian packaging and documents, both with character-level visibility ground truth, released with code, source lists and response-free results;
2. an error taxonomy grounded in visibility, which separates misreadings and fabrications of visible text from *completions* of text the camera did not capture, with prevalence per model and condition;
3. evidence, on synthetic and on real documents, that the models normalise unfamiliar printed words to familiar ones, and measurements of how image quality, framing and word category drive errors, with paired tests across models;
4. an evaluation of run-time detectors with operating points, costs and latencies, and an assessment of run-to-run stability.

2 Related work

2.1 AI visual assistance for blind and low-vision people

Remote sighted assistance and crowd-sourced answers [1] were followed by datasets of real photographs taken by blind people, which made their characteristics measurable: VizWiz images are frequently blurred, poorly framed or partly occluded, and many of them contain text [3, 14]. What blind and low-vision people want from image descriptions depends on the context, but transcribing visible text is consistently

valued [15]. Earlier computer-vision aids for visually impaired people focused on detection and recognition pipelines [5, 16]. Text readers built on such pipelines are now an established tool, although their accuracy degrades on real-world photographs [4].

Large multimodal models changed the interaction from reading out recognised text to conversational description. Diary and interview studies of apps such as Be My AI report broad adoption and new use cases, alongside errors that users find hard to detect and strategies for verifying output, such as re-taking photographs or asking again [6–8]. Studies of how blind people handle recognition errors show that verification is laborious and often impossible without sighted help [10, 11]. Blind users also tend to trust captions that are inconsistent with the context [9]. These findings motivate detection that works *inside* the assistive pipeline rather than relying on the user.

2.2 Hallucination in vision-language models

Object hallucination in image captioning was quantified early with the CHAIR metric, which attributed it partly to language priors [17]. For instruction-tuned VLMs it has been probed with polling questions in POPE [18] and with adversarial visual questions in HallusionBench [19]. Recent surveys organise the causes into data, model and decoding factors, and highlight the dominance of the language prior when visual evidence is weak [20, 21]. We use the same intuition but measure it on text, where the visual evidence can be degraded in a controlled way. Our earlier work on inclusive image captioning with large language models [22] examined description quality for visually impaired users. The present study isolates the text-reading component of such descriptions.

2.3 Reading text with vision-language models

Scene-text benchmarks such as TextVQA [23], ST-VQA [24] and TextCaps [25] established that answering and captioning often require reading. OCRBench showed that large multimodal models rely on semantic plausibility: they read meaningful words far better than non-semantic character strings [12]. MTVQA found a substantial accuracy gap for non-Latin scripts [13]. On food packaging, a controlled comparison under realistic print degradations found that vision-language models extract structured label information better than classical OCR, but that a small model exhibits a hallucination mode with high completeness and low accuracy [26]. Text in an image also strongly steers model behaviour, as typographic attacks show [27]. Closest to our work, recent studies document OCR hallucination by multimodal models on degraded documents [28] and fluent fabrication by VLM-based OCR on historical documents even where the character error rate is low [29]. None of these studies controls what is visible in the image or targets the assistive setting, Cyrillic script or Ukrainian orthography. We use synthetic rendering, a standard way to obtain exact text ground truth [30, 31], to provide this control.

2.4 Detecting hallucinations without ground truth

Black-box detectors exploit the observation that hallucinated content is unstable across samples: SelfCheckGPT compares a response with stochastic re-samples [32], and semantic entropy measures disagreement among sampled meanings [33, 34]. Models can also be asked to state their confidence [35, 36]. However, verbalised confidence tends to be overconfident [37], and hedging language does not faithfully reflect a model's intrinsic uncertainty [38]. For text reading, an independent OCR engine such as Tesseract [39] offers a cross-check that does not share the VLM's language prior. Our study compares these families of signals on the same quoted words.

3 Method

The study has two parts. Study 1 uses a fully controlled synthetic factorial design. Study 2 measures the same quantities on real Ukrainian packaging photographs and documents, and adds a near-word probe on real document pages. The models, prompts, labelling and detectors are shared.

3.1 Study 1: synthetic factorial stimuli

Content.

A generator written for this study produces Ukrainian text in three genres that blind users commonly need to read. *Signage* covers shop signs with opening hours, street-name plates, notices and public-transport stops. *Packaging* covers food labels with net weight, ingredients, best-before date and manufacturer, medicine boxes with dose, count, expiry date and batch number, and household chemicals with hazard warnings. *Documents* cover utility bills, shop receipts, medical referrals and building announcements. Content is assembled from curated Ukrainian vocabulary (e.g., 18 shop types, 30 real street names, 18 cities, real first names and patronymics) and random numerals (dates, times, prices, phone numbers, account and batch numbers). Each ground-truth token carries the category of the slot that produced it: *common word*, *real name*, *invented name*, *near-word name* or *digits*.

Every stimulus has one distinctive *name slot*: a brand, shop, company, street or patient surname, or a drug name. Its content is set by a balanced factor with three levels. In the *word* level it is a real word or name typical of Ukrainian brands (e.g., «Ластівка», «Барвінок») or a real surname or drug name. In the *invented* level it is a pronounceable string of Ukrainian syllables that is absent from a 290-thousand-word Ukrainian frequency list derived from OpenSubtitles[1] (e.g., «Фецедвус»). In the *near-word* level it is a real word with exactly one letter replaced by another letter of the same class, again checked to be absent from the word list (e.g., «Воложка» from «Волошка», «Метфермін» from «Метформін»). The near-word level is the sharpest probe of prior-driven normalisation: a model that reads what it expects rather than what is printed will "correct" it. Synthetic rendering gives complete control over these factors and an exact, independent ground truth. Its realism is limited, however, and

[1] https://github.com/hermitdave/FrequencyWords (2018 release, CC BY-SA 4.0).

Study 2 therefore repeats the measurements on real images (the two sets are compared in Section 4.5).

Rendering.

Lines are typeset on a panel styled for the genre: coloured sign plates with optional frames, pale product labels, or white and off-white paper, the latter with a monospaced font for receipts. Nineteen Cyrillic-capable font families are used, including serif, sans-serif, condensed, monospaced and decorative or handwritten faces; decorative faces appear only in titles. The panel is placed in a textured background scene through a random homography with small perspective distortion and rotation, so that it resembles a hand-held photograph. Because every character's box is known in panel coordinates, its position in the final image is known exactly.

Framing.

The second factor controls what reaches the camera. *Full frame* keeps the whole panel with a margin. *Crop* moves one frame edge (left, right or bottom) into the panel by 14–35%, which cuts lines and words as happens when a blind user aims a phone imprecisely [3]. *Occlusion* places either a skin-coloured finger entering from the bottom edge or a specular glare highlight over a random character of a random line. For each character we test a 3×3 grid of points inside its box. A character is *visible* if at least 8 of the 9 points are inside the frame and not occluded, *hidden* if at most 2 are, and *partial* otherwise. For glare, a point counts as occluded where the highlight's opacity exceeds 0.85. A token's state follows from its characters: visible, hidden, or partial if its characters are mixed. Crop and occlusion stimuli were re-drawn until at least one token was not fully visible, and full-frame stimuli until none was hidden.

Quality.

The third factor degrades the image after it is scaled to 1024 pixels on the long side. *Clean* adds mild sensor noise and JPEG compression at quality 90. *Blur* applies a Gaussian blur with $\sigma \in [1.6, 2.4]$ px, reduces contrast to 80–95% and compresses at quality 75. *Severe* applies $\sigma \in [2.8, 3.8]$ px, reduces contrast to 50–70% and brightness to 65–90%, adds stronger noise and compresses at quality 45. Small print becomes illegible under severe degradation while titles remain readable, which reproduces the graded legibility of real photographs taken by blind users.

Design.

The full factorial design crosses genre (3) $\times$ quality (3) $\times$ framing (3) $\times$ name-slot level (3), giving 81 cells with 7 images per cell, 567 images in total. The images contain 11 424 ground-truth tokens, of which 1 719 are not fully visible. The generator is seeded per stimulus, so the image set is reproducible byte for byte from the released code.

3.2 Study 2: real-image benchmark

All Study 2 stimuli are photographs or renderings of real Ukrainian objects and documents that a Ukrainian reader would recognise. Only the photographic conditions are

manipulated synthetically (framing and image quality, below), and a single word is edited in the near-word probe (Section 3.3). The benchmark has two genres, each with its own source of ground truth that is independent of the evaluated models. Signage could not be included in Study 2. We collected 70 freely licensed Wikimedia Commons photographs of Ukrainian signs, but the only available transcription of them was produced with a Claude model, a member of the evaluated family, so it is not independent of the systems under test. Signage is therefore covered by Study 1 only.

Packaging.

108 raw photographs of Ukrainian food packaging (ingredient panels and fronts) were taken from Open Food Facts, a collaborative database whose contributors photograph products in shops and at home. Next to each raw photograph, the database stores a Google Cloud Vision OCR result with a bounding box for every character, and this OCR is the ground truth. To make sure the OCR is reliable, an ingredient photograph is kept only if at least 60% of the words in the contributor's independent transcription of the ingredients occur in the OCR. A front photograph is kept only if at least half of the words of the contributor's product name occur in it. The contributor's transcription, product name and brand are added as further accepted reference text.

Documents.

71 real documents were collected as PDF files with a text layer from official and public Ukrainian sources. They include application forms of the Pension Fund, the Ministry of Justice, the State Land Cadastre and city councils' administrative-service centres (ЦНАП); Ministry of Health medical forms (e.g., 086/o, 027/o, 106/o); patient leaflets of medicines; electricity and gas bills and tariff notices from regional suppliers and the energy regulator; bank application forms and consumer-credit passports; university and school documents; announcements of councils and the National Police; and pages of library bulletins with book annotations, the closest available text to the back cover of a book. For each document, a block of consecutive lines with up to 150 words was cut from the first page (from the richest of pages 2–4 for library bulletins, whose first page is a cover). It was rendered at 150 dpi, and the text layer, with word boxes split into character boxes, is the ground truth. The rendered page is then "photographed" as a sheet lying on a textured surface, with a random homography (small perspective distortion and rotation) and uneven lighting. As a quality gate, Tesseract must find at least 60% of the text-layer words on the clean rendering. This rejected 3 documents, e.g., bilingual forms and PDFs whose text layer did not match the visible glyphs. 68 documents remain.

Framing.

Each stimulus is assigned one of three framing conditions. *Full frame* keeps the whole object. *Crop* moves one frame edge (left, right or bottom) into the text area by 14–35%, which cuts lines and words as happens when a blind user aims a phone imprecisely [3]. *Occlusion* places either a skin-coloured finger entering from the bottom edge or a specular glare highlight over a randomly chosen character. Because the position of every character is known, its visibility in the final frame can be computed. We test a

3×3 grid of points inside each character box. A character is *visible* if at least 8 of the 9 points are inside the frame and not occluded, *hidden* if at most 2 are, and *partial* otherwise. For glare, a point counts as occluded where the highlight's opacity exceeds 0.85. A token is visible, hidden, or partial if its characters are mixed.

Quality.

All stimuli are then scaled to 1024 pixels on the long side and assigned one of three quality levels. *Clean* adds only mild sensor noise. *Blur* applies a Gaussian blur with $\sigma \in [1.6, 2.4]$ px and reduces contrast to 80–95%. *Severe* applies $\sigma \in [2.8, 3.8]$ px, reduces contrast to 50–70% and brightness to 65–90%, and adds stronger noise. Images are stored as JPEG at the highest quality that keeps the file below 240 kB, because the model client silently drops larger image attachments. Within each genre, the nine (quality, framing) cells are balanced and assigned in a seeded random order.

Ground-truth categories.

Every ground-truth token is categorised as *digits*, a *common word* (present in a 290-thousand-word Ukrainian frequency list derived from OpenSubtitles[2]), or a *name or rare word* (absent from it: brands, surnames, place names, technical terms). The benchmark contains 176 images with 23 565 ground-truth tokens, of which 2 680 are not fully visible. Figure 1 shows examples.



Figure 1 Example Study 2 stimuli: real Ukrainian packaging photographs (Open Food Facts, top row) and official documents (bottom row) under different quality and framing conditions. The ground truth records every character and whether it is visible in the final frame. The benchmark also contains finger occlusions (23 of the 176 images), which are not shown here.

[2] https://github.com/hermitdave/FrequencyWords (2018 release, CC BY-SA 4.0).

3.3 Near-word probe

Prior-driven normalisation, where a model reports the word it expects rather than the word that is printed, cannot be measured on unedited text, because the printed and expected words coincide. We therefore edited the real document pages minimally. On each page, one common word of 6–12 letters that occurs once on the page was chosen at random. It was overpainted with a *near-word*: the same word with one interior letter replaced by another letter of the same class (vowel or consonant), checked to be absent from the word list (e.g., «документ» → «докуцент», «сплатити» → «сзлатити»). The near-word was typeset in a serif face at the original word's height and width, in the sampled ink colour. Each edited page was paired with its unedited original, and both were photographed with identical framing (full) and identical random degradation (clean or blur). Pages where the target word would be smaller than 11 px in the final image were skipped, leaving 64 pairs. In the description of each image we check whether the target is quoted in its printed form, whether the real word is quoted instead (normalisation), or whether some other form is quoted.

3.4 Models and prompts

We queried four production models of a single vendor, Anthropic, spanning its capability and price range: Claude Haiku 5.5, Sonnet 5.5, Opus 5.5 and Fable 5.1. All four were run on all 567 Study 1 images, all 176 Study 2 images and all probe pairs. The design is therefore a comparison *within one model family*, and none of our findings should be read as a statement about vision-language models in general. For cost reasons Fable 5.1, the most expensive model, received only the description prompt, and only descriptions were collected in the probe. To measure run-to-run stability, the description prompt was run twice more, on one Study 1 image per design cell (81 images), for every model. All calls used the vendor's command-line client in non-interactive mode with default sampling settings, no tools and a short system prompt in Ukrainian (all prompts are given verbatim in Appendix A): «Ти — сервіс опису зображень для незрячих і слабозорих користувачів. Відповідай українською мовою.» ("You are an image-description service for blind and low-vision users. Answer in Ukrainian."). Images were passed under opaque hashed file names so that the file name could not reveal the condition.

Two user prompts were used, each in a separate call without shared context. The *description* prompt imitates a blind user's request: «Я незряча людина. Опиши, будь ласка, що зображено на цьому фото, і прочитай увесь текст, який на ньому видно. Увесь прочитаний текст наводь дослівно в лапках «...».» ("I am a blind person. Please describe what is shown in this photo and read all the text that is visible on it. Give all the text you read verbatim in «...» quotation marks."). The request for guillemets makes the claimed text machine-extractable without an additional model. The *transcription* prompt asks for a verbatim line-by-line transcription, forbids corrections and completions, asks for a placeholder [нерозбірливо] ("illegible") where text is cut off, covered or unreadable, and asks for a "(?)" after any uncertain word. It serves both as a mitigation condition and as a run-time re-reading signal for detection.

3.5 Extracting and labelling claimed text

From each description we extract every guillemet-quoted span; an opening guillemet runs to the next closing one on the same line, so nested quotation marks cannot absorb descriptive prose. Straight and curly double quotes are also accepted. Bracketed or parenthesised remarks inside a quotation (e.g., «Осо[закрито пальцем] рахунок») are treated as annotations and removed, as are quotations that consist only of meta-words such as "covered" or of a single letter. Duplicate spans within a response are counted once. From each transcription, all lines are used after removing placeholders. Text is normalised by lower-casing, unifying apostrophe variants and mapping Latin look-alike letters to Cyrillic. The mapping is applied to the reference, the model output and the OCR alike, so mixed-script PDF text layers compare correctly. It is then split into letter and digit tokens. Digits separated by dots, colons or commas are merged, so dates, times and prices compare as units. Phone-like digit runs are also merged regardless of spacing, and leading zeros are dropped.

Each claimed token o receives one label from the ground truth of its image, including the accepted reference text:

- **correct**: o equals a visible token or a readable partial token, or o is an explicit fragment (at least two characters) of a visible token or a contiguous run of the visible characters of a partial token;
- **completion**: o equals the full form of a token at least one of whose characters is hidden, i.e., the text is true but the camera did not capture it;
- **misreading**: none of the above, but o is within a normalised edit similarity of 0.5 of some ground-truth token, i.e., printed text was rendered wrongly;
- **fabrication**: no ground-truth token is that similar, i.e., the text was invented.

A *hallucinated* token is a misreading or a fabrication; *unsupported* tokens additionally include completions. Misreadings are sub-typed as *digit* errors, *Ukrainian-letter* substitutions (the claimed and printed words differ only in і/и, ї/і, є/е, ґ/г or contain letters that exist only in Russian), *lexicalisations* (the claimed form is a real word or a known name but the printed one is not), and other substitutions. We also record whether the sentence containing a quotation carries a hedging cue (e.g., «ймовірно», «схоже», «нечітко», "probably", "looks like", "blurry"). Matching against the image's token set rather than a forced alignment is deliberately conservative: a wrong number that happens to occur elsewhere on the same page counts as correct. Conversely, text that is printed but absent from the reference (e.g., a word the reference OCR missed) is counted as a fabrication, so the audit below is essential.

Automated audit with two raters.

The labeller was audited by two independent automated raters, not by people. A stratified random sample of description tokens was drawn: 397 from Study 1 (25 per model and label) and 239 from Study 2 (15 per model and label, where available). Each rater received the stimulus image, the reference line the token was matched to, the quoted span, the token and the assigned label with its definition. It was asked (i) whether the label is right and (ii) whether the quoted word reproduces text that is printed and fully visible in the image. The second question is the one that matters for

the main metric. A misreading or fabrication is confirmed if the word is judged *not* printed-and-visible, a correct token if it is judged printed-and-visible, and a completion if it is judged not visible. The raters were Claude Sonnet 5.5 (through the same client as the evaluated runs) and the open model Gemma 3 12B (`gemma3:12b`, served locally by Ollama, temperature 0). Disagreements were resolved by a fixed conservative rule: an item counts as confirmed only if every rater that rated it confirms it. We report per-rater and resolved precision with Wilson intervals and Cohen's kappa [40] between the raters. This is agreement between automated raters, not human validation. One rater belongs to the evaluated model family, which may bias it towards the evaluated models' readings. The rater prompt is in Appendix A.

3.6 Detectors

For each token quoted in a description, five signals are computed without access to the ground truth:

1. *re-reading*: the token is absent from the same model's transcription of the same image; the variant *re-reading uncertainty* fires when the transcription marks the token with "(?)";
2. *cross-model*: the fraction of the other two main models whose description or transcription of the image does not contain the token;
3. *OCR*: one minus the best normalised edit similarity between the token and any token Tesseract 5.5 (`tessdata_best` Ukrainian model, page-segmentation modes 3 and 11 pooled) reads in the image [39];
4. *lexicon*: the token is not in the Ukrainian frequency list;
5. *hedging*: the quoting sentence carries a hedging cue;
6. *sample disagreement* (repeat subset only): the fraction of the model's two other descriptions of the same image that do not contain the token, a SelfCheckGPT-style signal [32].

Each signal is evaluated alone by the area under the ROC curve (AUROC) [41], and all signals are combined, together with the token's length and digit status, in a class-balanced logistic regression. Its scores are obtained by five-fold cross-validation with folds grouped by image, so that no image contributes to both training and testing. We report AUROC, average precision and, for deployment, the share of hallucinated tokens caught at fixed false-flag rates of 5, 10 and 20% on correct tokens. We also evaluate a simple rule that needs no training: flag a token if the model's re-reading lacks it, or if neither OCR nor a majority of the other models confirms it.

3.7 Statistics

Proportions are reported with Wilson 95% score intervals [42]. Because tokens are clustered within images, we additionally report percentile intervals from a cluster bootstrap over images [43] (1000 resamples for rates and 300 for AUROC). Every comparison stated in the text is tested. Comparisons on shared images (model versus model, description versus transcription) use the exact McNemar test on the image-level outcome "at least one hallucinated word", together with a paired cluster-bootstrap

95% interval (2000 resamples of images) for the difference in token-level hallucination rate. Comparisons between disjoint image groups (quality, framing, genre levels) use Fisher's exact test on the image-level outcome and an unpaired cluster-bootstrap interval for the token-rate difference. p-values are Holm-corrected within each family: the six model pairs per study, and the condition contrasts per study. All analysis code and the response-free aggregate results are released (see Declarations). Raw model responses are withheld from the repository but can be regenerated with the released scripts.

4 Results

All statements below concern four models of one vendor. Model-versus-model and condition-versus-condition differences are reported with the tests described in Section 3.7. All pairwise model tests are listed in Table B1 (Appendix B), and the full per-genre and per-condition breakdown is in Table B2 (Appendix B).

4.1 Study 1: how often the models assert text that is not there (RQ1)

Table 1 summarises the outcomes over all 567 synthetic images. Measured per quoted word, hallucination is rare: 3.8% of the words Haiku quotes in its descriptions are misread or fabricated, 2.8% for Sonnet, 3.2% for Opus and 4.1% for Fable. Measured per response, which is what a blind user experiences, it is common. A description contains at least one hallucinated word in 41% (Haiku), 35% (Sonnet), 38% (Opus) and 46% (Fable) of the images. Within this model family, the most expensive model is not the most faithful reader. Fable produces a hallucinated word in more descriptions than Sonnet (Holm-corrected McNemar $p =< 0.001$) and Opus ($p = 0.004$), and Haiku in more than Sonnet ($p = 0.009$). The other pairs do not differ reliably (Table B1). In 21–30% of the images, at least one hallucinated word is stated without any hedging in its sentence, so the listener has no cue to doubt it. Hallucination is not confined to degraded images: even for clean, fully framed images, 19% (Haiku), 11% (Sonnet) and 14% (Opus) of the descriptions contain a hallucinated word. At least one wrong digit group, which can affect dates, doses, prices, phone or account numbers, appears in 11%, 10% and 10% of the descriptions. Reading coverage is high: the models quote 92.7–95.8% of the visible ground-truth words.

Image quality and framing.

Table B2 (Appendix B) breaks down the rates by genre, quality, framing and word category. Figures in this paper show the real-image Study 2; the corresponding Study 1 numbers are given in the tables and the text. Severe degradation is the dominant driver. For every model, both the image-level rate and the token-level rate rise from clean to severe images (e.g., Haiku 70% vs 29% of descriptions with a hallucination, Holm $p =< 0.001$; token-rate difference [4.7, 7.2] percentage points; Sonnet 60% vs 23%, Opus 60% vs 25%, Fable 71% vs 32%, all $p < 0.001$). Moderate blur does not raise the rate: for every model the token-rate difference from clean includes zero (Haiku [-1.3, 0.1], Sonnet [-1.0, 0.3], Opus [-0.5, 0.9]). Occlusion by a finger or glare

Table 1 Study 1 (synthetic, 567 images): text-reading outcomes per model and prompt. Token rates are percentages of all quoted (description) or transcribed (transcription) tokens; brackets give Wilson 95% intervals. "Responses" is the percentage of images whose response contains at least one hallucinated (misread or fabricated) token. Coverage is the percentage of visible ground-truth tokens that the response reads correctly.

Model	Prompt	Tokens	Hallucinated %	Misread	Fabric.	Completion	Responses %	Cover.
Haiku 5.5	description	10,823	3.77 [3.43, 4.15]	2.86	0.91	1.89	41.1 [37.1, 45.2]	92.7
Haiku 5.5	transcription	10,576	4.37 [4.00, 4.77]	2.43	1.94	0.27	39.0 [35.0, 43.0]	94.7
Sonnet 5.5	description	11,458	2.82 [2.53, 3.14]	2.23	0.58	2.50	34.6 [30.8, 38.6]	95.6
Sonnet 5.5	transcription	10,806	3.75 [3.41, 4.12]	2.33	1.42	0.10	34.0 [30.3, 38.0]	96.8
Opus 5.5	description	12,146	3.18 [2.88, 3.51]	2.40	0.77	3.44	38.1 [34.2, 42.2]	95.8
Opus 5.5	transcription	10,691	2.12 [1.87, 2.41]	1.47	0.65	0.03	25.8 [22.3, 29.5]	97.6
Fable 5.1	description	12,709	4.10 [3.77, 4.46]	2.93	1.17	4.08	45.5 [41.4, 49.6]	95.8

raises the image-level rate for Haiku (53% vs 34%, $p = 0.009$) and Sonnet (46% vs 30%, $p = 0.061$), but not reliably for Opus ($p = 0.404$) or Fable ($p = 0.350$). A crop by the frame edge does not raise the misreading rate (all $p = 1$ after correction) but produces completions (Section 4.1). Per quoted word, signage is hallucinated more often than documents for every model (token-rate differences Haiku [1.9, 5.0], Sonnet [0.8, 3.2], Opus [1.7, 4.5] percentage points). The image-level difference, however, is not significant ($p = 1.000$ for Haiku), because signage carries fewer words per image.

What kind of text is misread.

The category of the printed word matters as much as image quality (Table B2). Common words are hallucinated at 2.0% (Haiku), 1.6% (Sonnet) and 1.5% (Opus), and real names at similar rates. Invented names, which no language prior can predict, are misread or replaced at 38.2%, 26.3% and 26.7%. Near-word names, one letter away from a real word, are misread at 29.2%, 19.0% and 29.6%. Digits sit in between (4.2%, 2.8%, 3.4%). For the near-word names, the misreading is very often the real word: 12.8% (Haiku), 9.7% (Sonnet) and 18.4% (Opus) of the claims matched to a near-word name are the lexicalised original. Examples are «Метфермін» read as «Метформін», «Добробїт» as «Добробут» and «Воложка» as «Волошка». The near-word probe on real documents (Section 4.4) shows the same normalisation on real pages.

Ukrainian orthography.

Among claims matched to invented names, 16.2% (Haiku), 13.0% (Sonnet) and 15.6% (Opus) differ from the printed form only in replacing ї, є or ґ by i, е or г, i.e., by spellings shared with Russian (e.g., «Піжєназол» read as «Піженазол»). This rate is inflated by the synthetic invented names, which over-represent these letters (Section 4.5). Across all hallucinated words in descriptions, the error taxonomy is as follows. Ukrainian-letter substitutions account for 13% (Haiku), 16% (Sonnet) and 17% (Opus), digit errors for 20%, 18% and 17%, lexicalisations for 8%, 8% and 17%, and fabrications for 24%, 21% and 24%. The remainder are other substitutions, such as grammatical "corrections", paraphrases inside quotation marks and wrong words in blurred small print.

Completions, hedging and the transcription prompt.

Crops and occlusions hide part of the text, and the description prompt often fills it in. Haiku produced 205 completions of partly or wholly hidden words, Sonnet 286 and Opus 418. The constrained transcription prompt, which explicitly forbids completion and offers a placeholder, nearly eliminates them (29, 11 and 3). Completions in descriptions are usually hedged (71%, 91% and 94%), whereas misreadings and fabrications mostly are not: 43% (Haiku), 38% (Sonnet) and 47% (Opus) of the hallucinated words are in a hedged sentence, compared with 11%, 12% and 13% of the correctly read words. This is consistent with reports that verbal hedging is only loosely tied to model uncertainty [38]. Whether the transcription prompt reduces misreadings depends on the model. Paired by image, it lowers the image-level rate for Opus (38% vs 26% of images, Holm $p =< 0.001$; token-rate difference [0.7, 1.4] percentage points), but not for Haiku ($p = 1.000$) or Sonnet ($p = 1.000$).

4.2 Study 1: detecting hallucinated words (RQ2)

Table 2 reports how well each run-time signal separates hallucinated from correct quoted words in descriptions. Agreement with the other two models is the strongest single signal (AUROC 0.90, 0.88 and 0.87 for Haiku, Sonnet and Opus). The model's own re-reading under the transcription prompt (0.77, 0.76, 0.78) and the Tesseract cross-check (0.77, 0.76, 0.74) follow. Tesseract reads Ukrainian poorly on degraded images, but it does not share the language prior of the evaluated models. Lexicon membership (0.67, 0.67, 0.63) and hedging cues (0.66, 0.62, 0.65) are weak. The lexicon fails because hallucinations are often real words. Because the peers in the cross-model signal come from the same vendor, this signal may be optimistic (Section 5).

The signals are complementary. The cross-validated combination reaches AUROC 0.969 [0.960, 0.977] for Haiku, 0.951 [0.937, 0.964] for Sonnet and 0.946 [0.933, 0.957] for Opus, against base rates of 3.8%, 2.8% and 3.2%. At a false-flag rate of 10% on correct words, it catches 91%, 86% and 83% of hallucinated words. A training-free rule flags a quoted word when the model's own re-reading lacks it, or when neither OCR nor a majority of the other models confirms it. It catches 78%, 68% and 69% of hallucinated words while flagging 4.7%, 5.3% and 5.9% of correct ones. The re-reading comparison alone catches 57–62%.

4.3 Study 2: real packaging and documents

Table 3 gives the outcomes on the 176 real images. Per quoted word, the descriptions are hallucinated at 5.5% (Haiku), 3.9% (Sonnet), 4.3% (Opus) and 5.6% (Fable). This is higher than in Study 1, and the response-level rates are much higher (58%, 64%, 63% and 73%), because a real ingredient panel or form page carries about six times as many words as a synthetic stimulus (Section 4.5). Coverage is lower (40.9–51.5%): descriptions of dense documents summarise rather than read everything. The model ordering is consistent with Study 1. Fable has a hallucinated word in more descriptions than Haiku (Holm $p = 0.001$), Sonnet ($p = 0.040$) and Opus ($p = 0.040$), and per quoted word Haiku is above Sonnet (difference 1.6 [0.8, 2.4] percentage points).

Table 2 Study 1: detection of hallucinated words in descriptions. AUROC with cluster-bootstrap 95% intervals (resampling images). The combined model is a class-balanced logistic regression over all signals plus token length and digit status, scored by five-fold cross-validation grouped by image. The rule row gives recall / false-flag rate.

Signal	Haiku 5.5	Sonnet 5.5	Opus 5.5
Re-reading (own transcription)	0.769 [0.74, 0.79]	0.758 [0.73, 0.79]	0.785 [0.76, 0.81]
Re-reading uncertainty mark (?)	0.555 [0.54, 0.57]	0.571 [0.55, 0.59]	0.554 [0.53, 0.58]
Cross-model agreement	0.905 [0.88, 0.92]	0.878 [0.85, 0.91]	0.869 [0.84, 0.89]
Tesseract OCR similarity	0.772 [0.75, 0.79]	0.757 [0.72, 0.79]	0.740 [0.71, 0.77]
Lexicon membership	0.672 [0.65, 0.69]	0.669 [0.64, 0.70]	0.632 [0.60, 0.67]
Hedging cue in sentence	0.656 [0.63, 0.68]	0.624 [0.60, 0.65]	0.654 [0.62, 0.68]
Combined (logistic, grouped 5-fold CV)	**0.969** [0.96, 0.98]	**0.951** [0.94, 0.96]	**0.946** [0.93, 0.96]
average precision (base rate)	0.717 (3.8%)	0.497 (2.8%)	0.393 (3.2%)
recall at 5% false flags	86.3%	76.2%	73.6%
recall at 10% false flags	90.9%	86.4%	83.4%
recall at 20% false flags	94.8%	93.2%	93.5%
Rule: not re-read, or unconfirmed by OCR and other models	77.9% / 4.7%	68.4% / 5.3%	69.2% / 5.9%
Tokens (hallucinated)	10,823 (408)	11,458 (323)	12,146 (386)

Packaging photographs are harder than documents for every model (token-rate difference for Opus [3.2, 6.1] percentage points, Holm $p = 0.044$ on the image level). Severe degradation raises the per-word rate (Haiku [3.7, 20.0], Sonnet [0.7, 7.7], Opus [0.8, 7.2] percentage points). The response-level rate, however, does not rise (all Holm $p \geq 0.7$), because the models quote far less text from severely degraded images. Words absent from the frequency list (names, brands and technical terms) are hallucinated at 5.5–9.2%, against 2.9–3.7% for common words and 6.0–7.2% for digits (Figure 4). Figure 3 shows the response-level rates per quality and framing cell, and Figure 5 the error taxonomy per model and genre. Figure 2 breaks the real material down by source. Front-of-pack photographs, dominated by brand names in decorative type, have by far the highest labelled rate. Part of this is a reference effect, because the OCR reference itself struggles with decorative type (Section 4.7). Among documents, bills and medical forms, with their dense numbers and abbreviations, are harder than plain notices, education documents and book annotations. On packaging, misreadings other than digits and Ukrainian letters and outright fabrications dominate. Completions are most frequent for Fable.

Table 3 Study 2 (real packaging and documents, 176 images): text-reading outcomes per model and prompt; columns as in Table 1.

Model	Prompt	Tokens	Hallucinated %	Misread	Fabric.	Completion	Responses %	Cover.
Haiku 5.5	description	8,428	5.47 [5.00, 5.98]	3.76	1.71	0.71	58.0 [50.6, 65.0]	40.9
Haiku 5.5	transcription	14,295	12.18 [11.65, 12.73]	5.88	6.30	0.45	76.7 [69.9, 82.3]	60.5
Sonnet 5.5	description	10,667	3.91 [3.56, 4.29]	2.72	1.19	0.53	63.6 [56.3, 70.4]	50.3
Sonnet 5.5	transcription	14,886	7.24 [6.84, 7.67]	3.71	3.53	0.15	75.6 [68.7, 81.3]	66.1
Opus 5.5	description	11,363	4.28 [3.92, 4.66]	2.85	1.43	1.01	63.1 [55.7, 69.8]	51.5
Opus 5.5	transcription	14,823	4.55 [4.23, 4.90]	2.73	1.83	0.13	69.3 [62.2, 75.7]	67.7
Fable 5.1	description	10,611	5.58 [5.16, 6.03]	3.68	1.90	2.23	73.3 [66.3, 79.3]	47.6

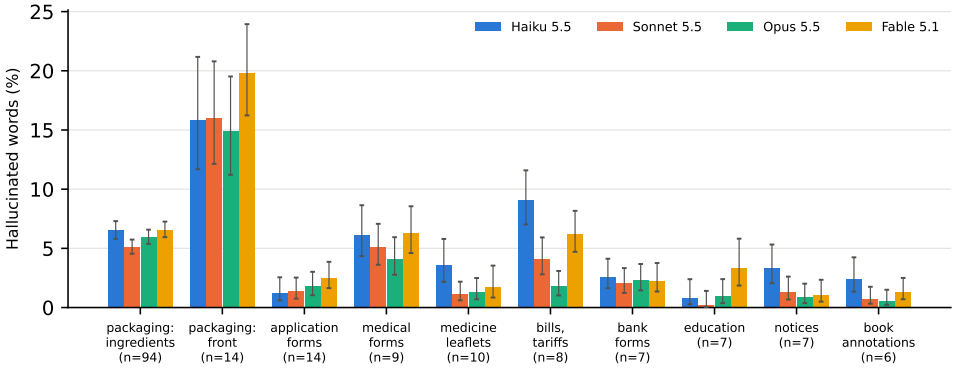


Figure 2 Study 2: token-level hallucination rate of descriptions by source category, per model, with Wilson 95% intervals; n is the number of images per category. Rates are before the audit adjustment.

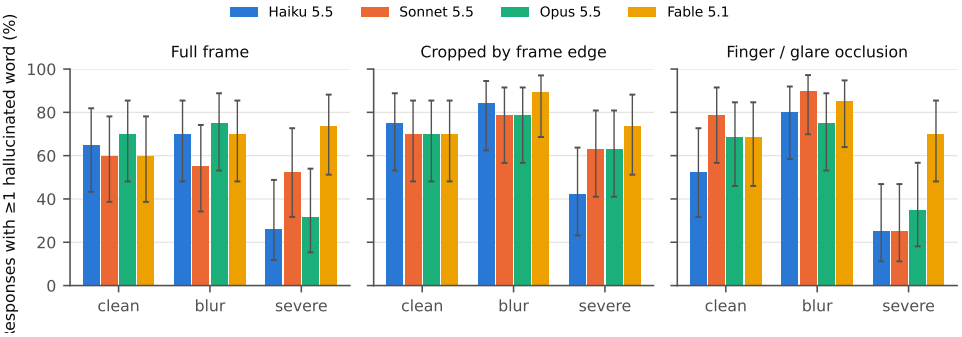


Figure 3 Study 2 (real images): share of description responses containing at least one hallucinated word, by image quality (x-axis) and framing (panels). Error bars are Wilson 95% intervals; each bar is based on 19 or 20 images, so the intervals are wide and no cell contrast survives Holm correction (Section 4.3).

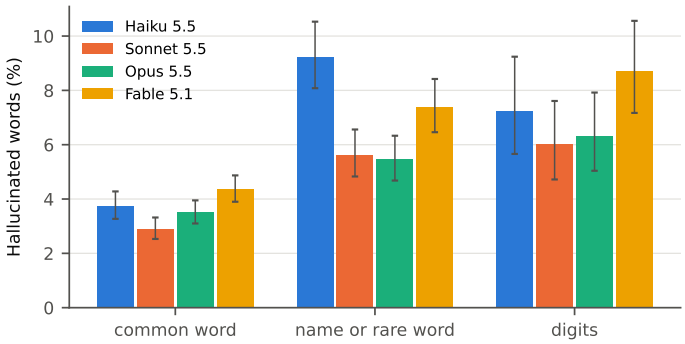


Figure 4 Study 2: share of quoted words that are hallucinated, by the category of the ground-truth word they correspond to (description prompt). Error bars are Wilson 95% intervals.

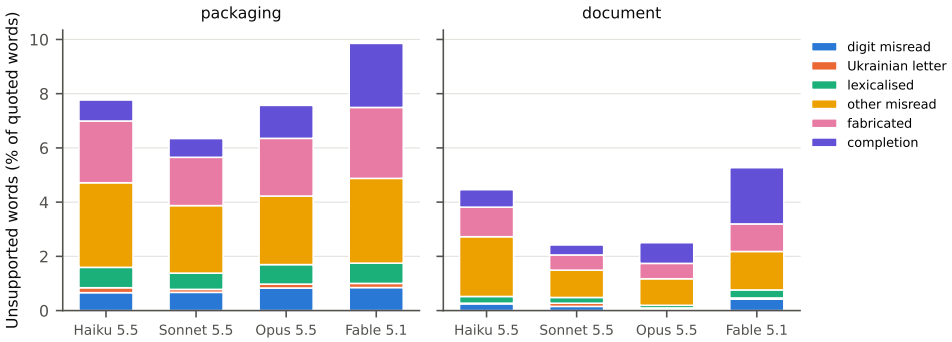


Figure 5 Study 2: error taxonomy per model and genre, as a share of all quoted words in descriptions. "Ukrainian letter" marks substitutions of і, ї, є, ґ by spellings shared with Russian; "lexicalised" marks a rare printed word reported as a common one. Rates are before the audit adjustment of Section 4.7.

Detection transfers to real images with lower but still useful performance (Table 4). The combined detector reaches AUROC 0.904 [0.867, 0.934] (Haiku), 0.863 [0.825, 0.899] (Sonnet) and 0.867 [0.837, 0.905] (Opus). At a 10% false-flag rate it catches 79%, 65% and 68% of hallucinated words. Cross-model agreement and OCR similarity remain the strongest single signals (Figure 6).

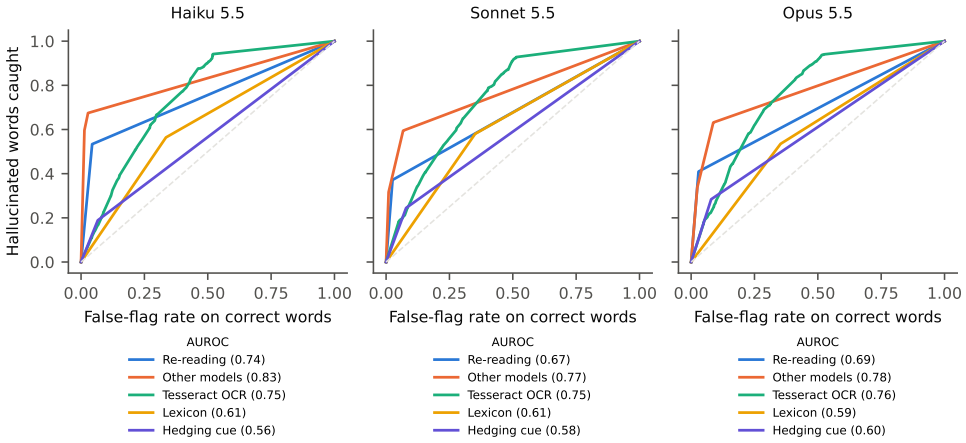


Figure 6 Study 2: ROC curves of the individual run-time signals for words quoted in descriptions of real images, per model. The legend gives each signal's AUROC.

Figure 7 shows whole descriptions of two real images with every quoted word marked by its label. It illustrates both the errors the benchmark is designed to catch and the main weakness of the Study 2 reference. On the yoghurt label, a surname

split across a line break is quoted as one corrected word and counted as a misreading. The other "hallucinations" are a barcode digit run, a stamp code and a superscript exponent, which are printed but missing from the OCR reference. On the employment-centre form, the abbreviation «МП» (seal place) is quoted as the expanded phrase «місце печатки», which is not printed. The run-time rule flags these words but also some correct ones. Figure 8 shows four further characteristic failures from the real-image runs.

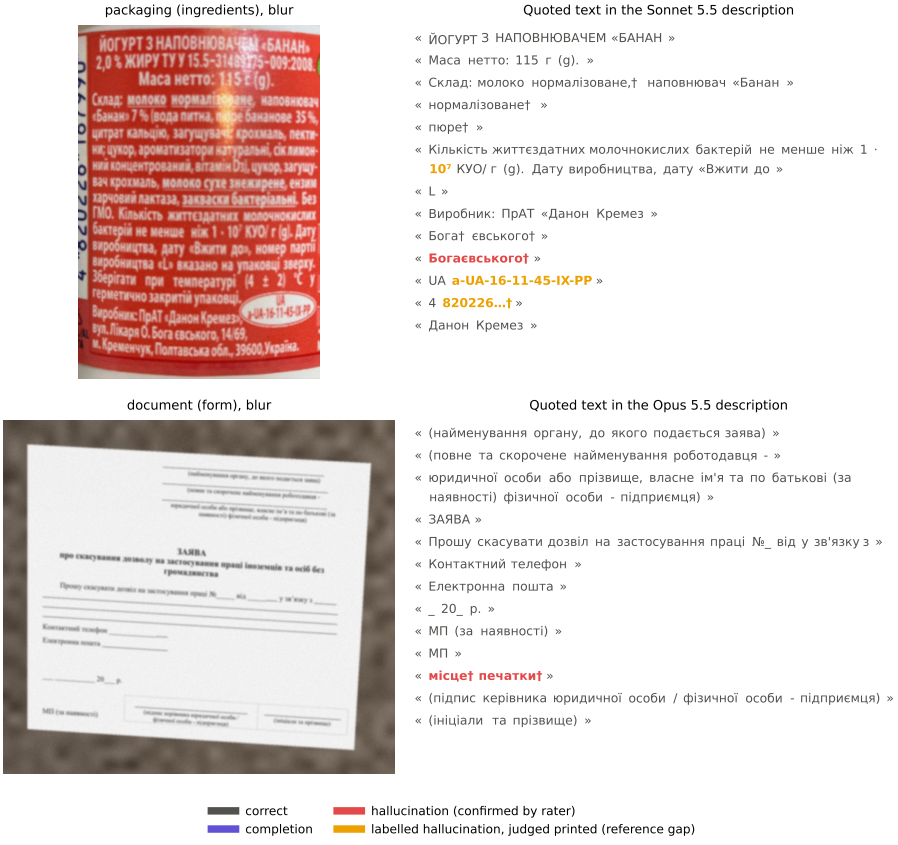
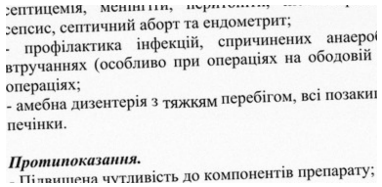


Figure 7 Two real stimuli with the quoted text of one model's description. Every quoted word is coloured by the deterministic label. Words labelled as hallucinated were additionally checked by the Claude audit rater: red where the rater confirms that the word is not printed, amber where it judges the word to be printed (a reference gap, e.g. a barcode or a stamp code missing from the OCR reference). A dagger marks words flagged by the training-free run-time rule. The two cases were selected for illustration; the rater verdicts are listed in the repository.

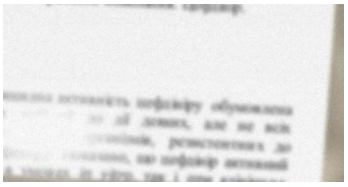
Table 4 Study 2: detection of hallucinated words in descriptions of real images; layout as in Table 2.

Signal	Haiku 5.5	Sonnet 5.5	Opus 5.5
Re-reading (own transcription)	0.745 [0.71, 0.77]	0.673 [0.64, 0.71]	0.690 [0.66, 0.73]
Re-reading uncertainty mark (?)	0.509 [0.50, 0.52]	0.522 [0.51, 0.54]	0.540 [0.52, 0.56]
Cross-model agreement	0.827 [0.79, 0.86]	0.771 [0.74, 0.81]	0.779 [0.74, 0.81]
Tesseract OCR similarity	0.749 [0.71, 0.78]	0.748 [0.71, 0.78]	0.765 [0.73, 0.81]
Lexicon membership	0.615 [0.59, 0.64]	0.614 [0.58, 0.64]	0.591 [0.57, 0.62]
Hedging cue in sentence	0.561 [0.54, 0.59]	0.583 [0.56, 0.61]	0.602 [0.57, 0.64]
Combined (logistic, grouped 5-fold CV)	**0.904** [0.87, 0.93]	**0.863** [0.83, 0.90]	**0.867** [0.84, 0.91]
average precision (base rate)	0.611 (5.5%)	0.365 (3.9%)	0.320 (4.3%)
recall at 5% false flags	75.3%	54.0%	49.4%
recall at 10% false flags	78.5%	65.0%	67.9%
recall at 20% false flags	84.0%	75.8%	77.6%
Rule: not re-read, or unconfirmed by OCR and other models	71.2% / 4.8%	46.3% / 3.0%	48.8% / 4.1%
Tokens (hallucinated)	8,428 (461)	10,667 (417)	11,363 (486)



Near-word normalised to the real word (edited document)

Printed: «тяжкям»
Sonnet 5.5 quoted: «тяжким»
Cause: language prior overrides a clearly printed one-letter difference.



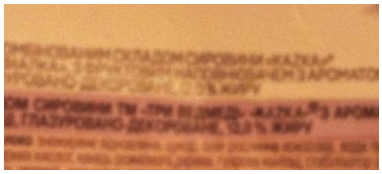
Ukrainian letter replaced by a Russian-like spelling

Printed: «...ми, бактерицидна активність цефдініру обумовлена»
Sonnet 5.5 quoted: «...ми, бактерицидна активність цефдиніру обумовлена пригніченням син...»
Cause: letters specific to Ukrainian (і, ї, є, ґ) are normalised away.



Completion of text cut off by the frame edge

Printed: «...не молоко, харчові волокна (олігофруктоза),»
Fable 5.1 quoted: «олігофруктоза»
Cause: the hidden part of the word is predictable, so it is filled in.



Digit misread

Printed: «...ЦІ, ГЛАЗУРОВАНО-ДЕКОРОВАНЕ, 12,0% ЖИРУ»
Opus 5.5 quoted: «Склад продукту: морозиво, 12,5 % жиру: молоко...»
Cause: small or blurred digits carry no language prior to correct them.

Figure 8 Failure cases from the real-image runs: crop of the stimulus around the token, printed text and quoted text. Top to bottom: a near-word on an edited medicine leaflet (probe, clean) reported as the real word; the Ukrainian spelling of a drug name (цефдініру, leaflet of 3-Dinir, blur with glare) quoted in its Russian-like form; a word cut off by the frame edge (packaging, severe, crop) quoted in full; a fat content of "12,0% on an ice-cream label quoted as "12,5% (packaging, blur with glare). Cases were drawn with a fixed seed among Sonnet, Opus and Fable descriptions.

4.4 Near-word probe on real documents

On the 64 probe pairs, the four models quote the unedited real word correctly on 52–70% of the original pages. On the edited pages they quote the printed near-word on only 3–30%, and on 44–56% they quote the real word instead (Haiku 48%, Sonnet 56%, Opus 44%, Fable 47%). For every model, the drop from reading the original correctly to reading the near-word correctly is significant (exact McNemar on the pairs: Haiku $p < 0.001$, Sonnet $p < 0.001$, Opus $p < 0.001$, Fable $p < 0.001$). The edit is clearly visible at the probe's resolution (the target letters are at least 11 px high). Prior-driven normalisation is therefore not an artefact of synthetic rendering: on real official documents, the models report the word they expect (Figure 9).

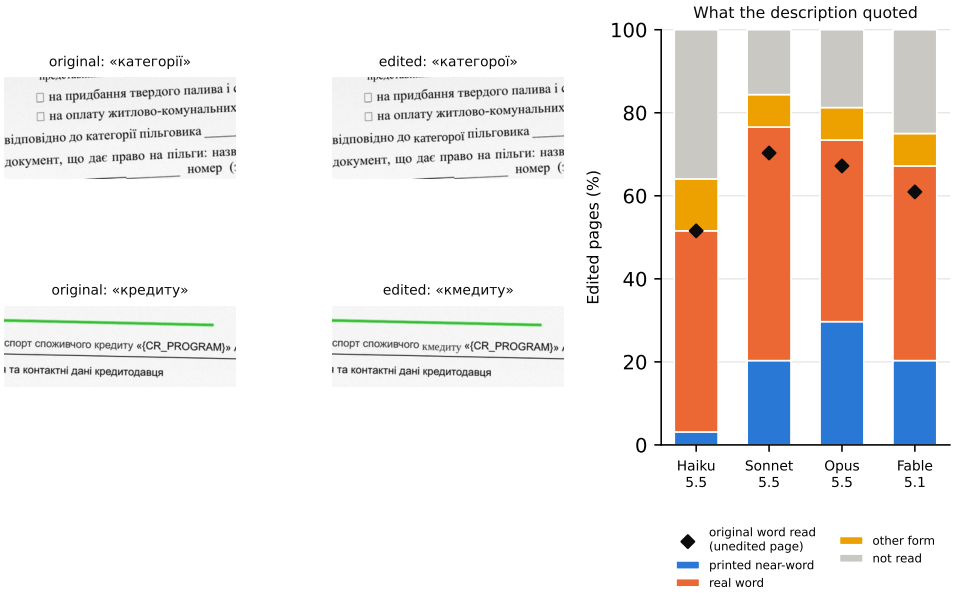


Figure 9 Near-word probe on real documents. Left: two of the 64 pairs (crop around the target on the original and the edited page, clean condition), drawn with a fixed seed among pages where at least three models reported the real word. Right: what each model's description quoted for the target on the edited pages, and how often it read the target correctly on the unedited original (diamonds).

4.5 Synthetic versus real stimuli

The two sets differ in ways that matter for interpretation. Real stimuli carry 122 ground-truth tokens per image against 20 in Study 1, with similar word length (mean 6.0 versus 5.7 letters). Names and rare words make up 26% of real tokens against 9% of synthetic ones, and digits 10% against 23%. Study 1 thus under-represents rare words and over-represents digits. The invented names of Study 1 over-represent the letters

specific to Ukrainian relative to natural Ukrainian text (OpenSubtitles frequency list): ї by a factor of 13, є by 12 and ґ by 93. In the real texts, ґ is almost absent (0.00% of letters). The Ukrainian-letter error rate of Study 1 should therefore be read as a stress test, not as an estimate for everyday text. In Study 2, Ukrainian-letter substitutions make up only 2% (Haiku), 3% (Sonnet), 2% (Opus) and 2% (Fable) of the hallucinated words, although they do occur on real leaflets (Figure 8). Lexicalisations make up 9–11% and fabrications 30–34%. The main effects that Study 1 identified are reproduced on real images: hallucinated words in a majority of responses, unhedged errors, normalisation of near-words, degradation-driven misreading, and a usable combined detector. Their magnitudes differ.

4.6 Run-to-run stability

The description prompt was run three times on 81 Study 1 images, one per design cell (Table 5). The image-level outcome (any hallucinated word) changes between runs for 19–25% of the images. The token-level rate varies by at most 1.28 percentage points between runs. Point estimates of individual responses are therefore noisy, while the aggregate rates are stable. Disagreement between a model's own samples is a usable detector in its own right (AUROC 0.76–0.81 on this subset). This is comparable to, and for some models slightly above, the model's re-reading under the transcription prompt on the full set.

Table 5 Run-to-run stability of the description prompt on 81 Study 1 images (one per design cell), three runs per model. "Flip" is the share of images whose image-level outcome (any hallucinated word) is not the same in all three runs, with a Wilson 95% interval. "Disagr." is the AUROC of sample disagreement as a detector of hallucinated words in run 1.

Model	Images	Any-hall. % (r1/r2/r3)	Flip %	Token rate % (r1/r2/r3)	Spread	Disagr.
Haiku 5.5	81	42 / 36 / 37	19 [12, 28]	3.42 / 3.29 / 3.27	0.14	0.77
Sonnet 5.5	81	36 / 36 / 37	25 [17, 35]	2.84 / 3.21 / 3.42	0.58	0.81
Opus 5.5	81	31 / 36 / 36	20 [13, 30]	2.69 / 3.02 / 2.64	0.38	0.76
Fable 5.1	81	38 / 40 / 48	25 [17, 35]	3.25 / 3.52 / 4.53	1.28	0.77

4.7 Automated audit of the labeller

On the Study 1 sample, the Claude rater confirms 98% of correct labels, 93% of misreadings, 84% of fabrications and 99% of completions. On the Study 2 sample the corresponding figures are 97%, 58%, 54% and 83%. Gemma 3 12B rated 162 of the Study 1 items before its server became unavailable. It confirms correct labels at a similar rate (98%) but judges most misread words to be printed as they were quoted (13% confirmed). Inspection of its reasons shows that it does not resolve single-letter differences in Ukrainian words. Agreement between the raters on the printed-and-visible question is accordingly low (Cohen's $\kappa = 0.22$ on 162 items, raw agreement 54%). Under the conservative resolution rule, 73% (66–78%) of the Study 1 hallucination labels are confirmed. The Claude rater alone confirms 88%. For Study 2 the Claude rater

confirms 56% of hallucination labels. The difference reflects reference gaps on real material, e.g., logos and stamps that are drawn as images rather than text and words that the reference OCR missed. Reported hallucination rates should therefore be read as upper bounds. Scaling the token-level rates of the descriptions by the Claude rater's confirmation rate gives audit-adjusted rates of 2.5–3.6% for Study 1 and 2.2–3.1% for Study 2. The large correction on real material is the main caveat on the Study 2 numbers. Gemma 3 was not available for the Study 2 sample, so these items have a single rater. One rater is a member of the evaluated family, and the other rater's own reading of Ukrainian is the bottleneck, so neither replaces a human validation.

4.8 Cost and latency

Table 6 reports tokens, list-price cost and wall-clock time per image. All values were measured through the same command-line client, from the same machine and on the same days, so they are comparable across the four models but include client overhead and network time. On real images, a Haiku description costs 0.0012 USD and takes about 10 s. The re-reading call that the strongest single-model detector needs (a transcription by the same model) adds 0.0012 USD and 8 s for Haiku, and 0.0403 USD and 11 s for Opus. The cross-model signal requires two further descriptions, i.e., at least two to three times the cost and latency of a single description.

Table 6 Cost and latency per image and call, by study, model and prompt: mean input tokens (including cached system-prompt tokens), mean output tokens, mean list-price cost in USD, and median (90th percentile) wall-clock seconds. All calls used the same command-line client path.

Study	Model	Prompt	Calls	In tok.	Out tok.	USD	Seconds
Study 1	Haiku 5.5	description	567	1,728	692	0.0007	6.0 (7.8)
Study 1	Haiku 5.5	transcription	567	1,941	492	0.0006	5.0 (7.8)
Study 1	Sonnet 5.5	description	567	1,817	809	0.0153	9.4 (12.5)
Study 1	Sonnet 5.5	transcription	567	1,962	125	0.0090	5.6 (9.3)
Study 1	Opus 5.5	description	567	1,889	738	0.0298	11.6 (16.7)
Study 1	Opus 5.5	transcription	567	1,939	276	0.0210	5.8 (8.3)
Study 1	Fable 5.1	description	567	1,880	659	0.0705	10.7 (16.7)
Study 2	Haiku 5.5	description	176	3,304	1,514	0.0012	9.6 (12.9)
Study 2	Haiku 5.5	transcription	176	3,405	1,524	0.0012	8.5 (12.2)
Study 2	Sonnet 5.5	description	176	3,146	1,486	0.0239	13.9 (18.1)
Study 2	Sonnet 5.5	transcription	176	3,372	821	0.0177	6.7 (12.0)
Study 2	Opus 5.5	description	176	3,288	1,565	0.0497	17.9 (24.1)
Study 2	Opus 5.5	transcription	176	3,472	1,056	0.0403	11.2 (18.2)
Study 2	Fable 5.1	description	176	3,677	1,319	0.1148	17.5 (29.3)

5 Discussion

5.1 What the error profile means for blind users

The headline token-level rates of a few per cent understate the accessibility problem. A blind user does not receive tokens but descriptions. Between 35–46% of the synthetic descriptions and 58–73% of the descriptions of real packaging and documents

contained at least one word that is not printed or not visible. Most of these words were delivered without any cue of uncertainty. The errors also concentrate where they hurt. Names and rare words, which a language prior cannot predict, are hallucinated more often than common words in both studies, and they are what a user needs to tell one medicine, bill or product from another. Near-words are reported as the real word they resemble, on synthetic labels and on real official documents alike. A user told that a leaflet says "тяжким" has no way to learn that it says something else. The rare case in which an unusual printed string matters is the case the models are least likely to report faithfully.

Two further patterns are specific to the Ukrainian setting. First, letters that distinguish Ukrainian from Russian orthography are sometimes replaced by spellings shared with Russian. This happens on real medicine leaflets too, although less often than our stress test with synthetic invented names suggests. The change is easy to miss in a screen-reader rendering, but it alters the spelling of names that users may have to repeat, type or search for. Second, the models describe text that is cut off by the frame edge or covered by a finger by completing it. Completions are usually hedged and often right, but they are not readings. Because aiming errors are frequent in photographs taken by blind people [3], a description that fills in the unseen part removes the cue that would prompt the user to re-take the photograph.

Within the one model family we tested, capability is no remedy. The most expensive model, Fable 5.1, produced a hallucinated word in significantly more descriptions than the mid-range models in both studies. It tends to say more and to complete more of what is hidden, and all four models share the prior-driven normalisation of unfamiliar words. Whether the same holds for other vendors' models is open. Within this family, choosing a larger model does not replace a run-time check.

5.2 Design recommendations

The results suggest the following measures for assistive applications that read text with models of this kind:

1. *Re-read before reading out.* A second, constrained transcription call by the same model is the cheapest single check. For Haiku it adds 0.0012 USD and about 8 s per real image (Table 6). Comparing the two readings word by word catches 57–62% of the hallucinated words in Study 1. Adding an on-device OCR pass, which shares no language prior with the model, and a second model where available raises this to 68–78% at a false-flag rate of 5–6%. On real images the gains are smaller (Table 4), and a second model at least doubles cost and latency.
2. *Mark, do not hide.* Flagged words should be voiced with an explicit marker (e.g., "probably ...") or spelled out letter by letter, rather than removed. This lets the user decide whether to re-take the photograph or seek sighted verification, in line with the verification strategies blind users already employ [7, 10].
3. *Separate reading from inference.* Prompts and interfaces should distinguish "the label says" from "the hidden part is probably". Our transcription prompt shows that completions can be almost eliminated by instruction. The description prompt should therefore ask the model to state explicitly where the frame cuts the text, which also tells the user how to move the camera.

4. *Treat names and numbers as critical.* Digits and names should always pass the cross-check and, when flagged, be offered for character-level reading.

These recommendations follow from measured detector performance, not from tests with users. Whether they improve decisions is a question for a user study (below).

5.3 Limitations

One vendor. All evaluated models are Claude models of one vendor, queried through its command-line client with default settings. No statement in this paper is about vision-language models in general. Other model families, decoding settings and on-device models may behave differently. In particular, the cross-model detector may be optimistic here, because the peer models share a vendor and plausibly some of their errors. With independent vendors' models it may perform worse or better. A multi-vendor replication on the released stimuli is the most important next step. Fable 5.1 received only the description prompt, so it is not part of the detection analysis.

Synthetic versus real stimuli. Study 1 is fully synthetic. Section 4.5 quantifies how it differs from real text: about six times fewer words per image, fewer rare words, more digits, and invented names that over-represent ї, є and especially ґ relative to natural Ukrainian. Its absolute rates, in particular the Ukrainian-letter error rate, are therefore not estimates for everyday text. Study 2 replaces the content with real packaging and documents, but its degradations (blur, cropping, finger and glare) are still applied synthetically. Its documents are renderings of PDF pages laid on a surface, without folds, curvature or real lighting. Its packaging photographs were taken by sighted contributors and are better framed than typical photographs taken by blind people [3]. The near-word edit is typeset in a generic serif face, and an attentive reader may notice it.

Ground truth. Each Study 2 genre has its own reference: Google Cloud Vision OCR checked against an independent contributor transcription for packaging, and the PDF text layer checked against Tesseract for documents. Both can miss text that is drawn rather than typeset (logos, stamps) or misread decorative type, and a model that correctly reads such text is scored as fabricating. Signage photographs were collected but excluded, because no reference independent of the evaluated model family was available. The automated audit indicates how large the resulting overstatement is.

Labelling and audit. Claimed text is identified by quotation marks, which the prompt requested. Text a model reads out without quoting it is not scored, and quoted text that a model presents as an alternative reading is scored as a claim. The audit used two automated raters, not people. One rater belongs to the evaluated family, and the other, an open 12-billion-parameter model, could not resolve single-letter differences in Ukrainian and agreed poorly with the first. A human audit by Ukrainian readers remains necessary for a validated error rate.

Stability and scale. Individual responses vary between runs (Section 4.6), so per-image results are noisy. Aggregate rates are stable, but the run-to-run check covered only 81 images. Study 2 has fewer than 20 images per model in each (quality, framing, genre) cell, so its condition-level comparisons are underpowered and mostly not significant after correction.

No user study. We measure what is said, not how blind users perceive or act on it. Hallucination rates are proxies for the risk of acting on wrong information, not measures of harm. How flagged words should be voiced, and whether flags improve decisions without eroding trust in correct readings, requires a study with blind participants using screen readers in realistic reading tasks.

6 Conclusion

Four Claude models read Ukrainian signs, labels and documents well enough that per-word error rates are low. They nevertheless assert text that is not printed in a large share of their descriptions: in 35–46% of descriptions of synthetic stimuli and 58–73% of descriptions of real packaging and documents. The main mechanisms are normalising unusual words to familiar ones, which we reproduced on real official documents with a one-letter edit, altering digits, completing text the camera did not capture and, less often on real text, replacing Ukrainian-specific letters. Most of these assertions carry no hedge, and the most capable model of the family is not the most faithful reader. The errors are detectable at run time. A second constrained reading, an independent OCR pass and agreement between models together separate hallucinated from correct words with an AUROC of 0.95–0.97 on synthetic and 0.86–0.90 on real images. Assistive applications can therefore tell blind users *which* words to doubt. Whether these findings hold for other vendors' models, and whether such flags help users, remains to be tested. The stimulus generator, source lists, ground truth and analysis code are released to support both.

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Competing interests. The authors have no competing interests to declare that are relevant to the content of this article.

Ethics approval and consent to participate. Not applicable. The study involved no human participants. It used synthetic images, publicly available Open Food Facts packaging photographs (CC BY-SA 3.0) and documents publicly posted by official and public Ukrainian sources, which were processed only by automated model and OCR queries.

Consent for publication. Not applicable.

Data availability. The complete ground truth of both studies (every character, its category and its visibility), the source URL and licence of every Study 2 image (Open Food Facts photographs: CC BY-SA 3.0; documents: official or publicly posted documents, see the manifest), the near-word probe definitions and the response-free aggregate results are available in the project repository, https://github.com/rzats/

universal-access-in-2026. The Study 1 images are regenerated byte for byte by the released generator. Study 2 images and raw model responses are not redistributed; they can be regenerated from the listed sources with the released scripts, subject to the sources' licences and the model provider's terms.

Materials availability. Not applicable.

Code availability. The stimulus generator, model-query, OCR, labelling, analysis and figure scripts are available in the same repository.

Author contribution. Rostyslav Zatserkovnyi: conceptualisation, study direction and supervision, review of methods and results, writing (review and editing). Roksoliana Zatserkovna: conceptualisation, accessibility domain expertise, review of results, writing (review and editing). Petro Kutsyk: review of the manuscript. The experiment code, model runs, analyses and the labeller audit were automated with an AI system (see Use of generative AI). All authors read and approved the final manuscript.

Use of generative AI. During the preparation of this work the authors used Claude (Anthropic) to proofread the manuscript and check grammar and spelling. In addition, Claude was used under the authors' direction to implement the stimulus generation, model querying, labelling and analysis code, to run the experiments and the automated labeller audit, and to draft the text, which the authors reviewed. The Claude models evaluated as the objects of study are described in Section 3.4.

Appendix A Prompts, models and run settings

Evaluated models and client.

`claude-haiku-5-5`, `claude-sonnet-5-5`, `claude-opus-5-5` and `claude-fable-5-1` (Anthropic). Every call went through the Claude Code command-line client, version 2.1.304, in non-interactive mode (`claude -p … --output-format json --system-prompt … --tools "" --setting-sources "" --no-session-persistence`). The image was attached by an `@`-reference to a file with an opaque hashed name. The client does not expose temperature or top-p, so the provider's default sampling settings apply. Each call was a fresh session without shared context. Study 1 and the Study 2 signage and document runs were made on 21–22 October 2026. The Study 2 packaging runs, the remaining Opus and Fable calls, the near-word probe, the repeated runs and the audit were made on 28–29 October 2026. Token counts, list-price cost and wall-clock time of every call were recorded from the client's JSON output (Table 6).

System prompt (all evaluated calls).

> Ти — сервіс опису зображень для незрячих і слабозорих користувачів. Відповідай українською мовою.

Description prompt.

The attachment reference `@<file>.jpg` follows on a new line.

Я незряча людина. Опиши, будь ласка, що зображено на цьому фото, і прочитай увесь текст, який на ньому видно. Увесь прочитаний текст наводь дослівно в лапках «...».

Transcription prompt.

Перепиши дослівно весь текст, який видно на цьому зображенні, рядок за рядком. Нічого не виправляй і не доповнюй. Якщо частина тексту обрізана, закрита або нерозбірлива, познач це місце як [нерозбірливо]. Якщо ти не впевнений у слові, познач його знаком (?) одразу після слова. Виведи лише текст.

Audit rater prompt (both raters, identical).

The placeholders in braces are filled per token. The stimulus image is attached: for Claude through the same client as above with model `claude-sonnet-5-5`, for Gemma as a base64 image in an Ollama `/api/chat` request.

You are auditing an automatic scorer of image descriptions written for blind users. A vision-language model described the attached image and quoted text it claimed to read. The automatic scorer compared one quoted word with the reference text of the image and assigned a label.
Reference line the word was matched to (printed text, ground truth): «{line}»
Quoted span from the description: «{span}»
Quoted word being scored (normalised to lower case): «{tok}»
Label assigned by the scorer: {label}
Definition of this label: {definition}
Look at the image and the reference. Is the assigned label right for this word? Ignore capitalisation, punctuation and spacing differences. Answer with JSON only: {"verdict": "right" or "wrong", "reason": "<at most 15 words>"}

The label definitions inserted into the prompt are as follows. *Correct*: the token matches text that is printed and visible in the image (an explicit fragment of a visible word also counts). *Completion*: the token matches a printed word that is partly or wholly cut off by the frame edge or covered by a finger or glare, so the model completed text it could not see. *Misread*: the token is a wrong reading of a printed word (wrong letters or digits, a different word form, a Russian-like spelling, a "corrected" spelling). *Fabricated*: the token does not correspond to any printed text in the image; the model invented it.

Open audit rater.

`gemma3:12b` (Ollama digest `f4031aab637d`, Q4_K_M quantisation) served by Ollama 0.40.11 on a separate local machine, one request at a time. Decoding used temperature 0, seed 7 and at most 120 output tokens. Its latency is not comparable with the cloud models and is not reported.

Appendix B Pairwise model tests and per-condition breakdown

Table B1 lists every pairwise comparison between the four models on shared images, for both studies, with Holm-corrected exact McNemar tests on the image-level outcome

Table B1 Pairwise model comparisons on shared images (description prompt). "Any-hall." gives the share of images with at least one hallucinated word for the first / second model; "Disc." the discordant pairs (first only / second only); p is the exact McNemar test, Holm-corrected across the six pairs of each study; the last column is the difference in token-level hallucination rate (first minus second, percentage points) with a paired cluster-bootstrap 95% interval.

Pair	Images	Any-hall. %	Disc.	p_{Holm}	Δ token rate [95% CI]
Study 1 (synthetic)					
Haiku 5.5 vs Sonnet 5.5	567	41.1 / 34.6	89/52	0.009	0.95 [0.50, 1.43]
Haiku 5.5 vs Opus 5.5	567	41.1 / 38.1	95/78	0.224	0.59 [0.07, 1.10]
Haiku 5.5 vs Fable 5.1	567	41.1 / 45.5	73/98	0.198	-0.33 [-0.84, 0.17]
Sonnet 5.5 vs Opus 5.5	567	34.6 / 38.1	51/71	0.198	-0.36 [-0.74, 0.04]
Sonnet 5.5 vs Fable 5.1	567	34.6 / 45.5	49/111	<0.001	-1.28 [-1.81, -0.81]
Opus 5.5 vs Fable 5.1	567	38.1 / 45.5	56/98	0.004	-0.92 [-1.42, -0.43]
Study 2 (real)					
Haiku 5.5 vs Sonnet 5.5	176	58.0 / 63.6	17/27	0.523	1.56 [0.80, 2.42]
Haiku 5.5 vs Opus 5.5	176	58.0 / 63.1	14/23	0.523	1.19 [0.25, 2.40]
Haiku 5.5 vs Fable 5.1	176	58.0 / 73.3	12/39	0.001	-0.11 [-1.03, 0.90]
Sonnet 5.5 vs Opus 5.5	176	63.6 / 63.1	19/18	1.000	-0.37 [-0.98, 0.14]
Sonnet 5.5 vs Fable 5.1	176	63.6 / 73.3	11/28	0.040	-1.67 [-2.34, -1.03]
Opus 5.5 vs Fable 5.1	176	63.1 / 73.3	12/30	0.040	-1.30 [-1.96, -0.62]

and paired cluster-bootstrap intervals for the token-rate difference. Table B2 gives the token-level hallucination rate of descriptions for every level of every factor, so that no result in the main text is reported only in pooled form.

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Table B2 Token-level hallucination rate (%) of descriptions by study, factor and level, with the number of quoted tokens in parentheses. Study 1: synthetic, Study 2: real images. Signage in Study 2 is used in full frame only. The invented and near-word categories exist only in Study 1. "Name / rare word" in Study 2 covers every word absent from the frequency list.

Study	Factor	Level	Haiku 5.5	Sonnet 5.5	Opus 5.5	Fable 5.1
1	genre	signage	6.5 (1,971)	4.4 (2,166)	5.6 (2,324)	6.5 (2,306)
1	genre	packaging	3.2 (4,101)	2.4 (4,343)	2.6 (4,568)	3.4 (4,830)
1	genre	document	3.1 (4,751)	2.4 (4,949)	2.6 (5,254)	3.8 (5,573)
1	quality	clean	2.1 (3,789)	1.7 (3,893)	1.7 (4,182)	2.2 (4,341)
1	quality	blur	1.6 (3,713)	1.3 (3,873)	1.9 (4,046)	1.9 (4,221)
1	quality	severe	8.1 (3,321)	5.6 (3,692)	6.1 (3,918)	8.3 (4,147)
1	framing	full frame	3.0 (3,521)	2.6 (3,677)	3.1 (3,743)	3.9 (3,840)
1	framing	crop	3.4 (3,418)	2.2 (3,573)	2.6 (3,898)	3.7 (4,093)
1	framing	occlusion	4.8 (3,884)	3.6 (4,208)	3.8 (4,505)	4.6 (4,776)
1	category	common word	2.0 (7,486)	1.6 (7,918)	1.5 (8,342)	1.6 (8,643)
1	category	name / rare word	3.4 (530)	2.7 (556)	1.7 (588)	3.1 (611)
1	category	invented name	38.2 (217)	26.3 (251)	26.7 (266)	36.9 (314)
1	category	near-word name	29.2 (202)	19.0 (232)	29.6 (297)	32.0 (309)
1	category	digits	4.2 (2,388)	2.8 (2,501)	3.4 (2,653)	5.3 (2,832)
2	genre	packaging	7.0 (4,391)	5.7 (5,501)	6.3 (6,250)	7.5 (5,886)
2	genre	document	3.8 (4,037)	2.1 (5,166)	1.7 (5,113)	3.2 (4,725)
2	quality	clean	4.2 (3,953)	3.7 (5,149)	3.9 (5,453)	4.8 (4,764)
2	quality	blur	5.1 (3,848)	3.5 (4,682)	4.1 (5,074)	5.4 (4,595)
2	quality	severe	15.3 (627)	7.7 (836)	7.5 (836)	9.0 (1,252)
2	framing	full frame	5.7 (2,728)	3.7 (3,544)	4.7 (3,835)	6.6 (3,612)
2	framing	crop	6.0 (2,860)	4.3 (3,505)	4.1 (3,740)	5.1 (3,658)
2	framing	occlusion	4.7 (2,840)	3.7 (3,618)	4.0 (3,788)	5.0 (3,341)
2	category	common word	3.7 (5,480)	2.9 (6,866)	3.5 (7,339)	4.4 (6,807)
2	category	name / rare word	9.2 (2,134)	5.6 (2,752)	5.5 (2,918)	7.4 (2,737)
2	category	digits	7.2 (814)	6.0 (1,049)	6.3 (1,106)	8.7 (1,067)

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